Lesson Planning

Complete Lesson Plan Book

Class: Class 9

Subject: Physics

Medium: English

Academic Year 2026-27

DM

This book is prepared as a teaching guide for teachers.

Table of Contents

Unit 1: Speed, Velocity and Acceleration

DM

Unit 1: Speed, Velocity and Acceleration

Learning Objectives

• Students will be able to define speed, velocity and acceleration in their own words.

• Students will be able to calculate average speed using v = d/t and acceleration using a = (vf - vi)/t.

• Students will be able to distinguish scalar quantities (speed) from vector quantities (velocity, acceleration).

• Students will be able to interpret the sign of velocity and acceleration for motion in a straight line.

Key Concepts

• Speed is distance covered per unit time (a scalar): average speed = total distance / total time, unit m/s.

• Velocity is displacement per unit time and has direction (a vector); a car turning at constant speed changes velocity.

• Acceleration is the rate of change of velocity: a = (final velocity - initial velocity) / time, unit m/s².

• Positive acceleration means speeding up in the chosen direction; negative acceleration (retardation) means slowing down.

Teaching Methodology (Step by Step)

1. Start with the question: a cyclist rides 60 km in 2 hours — how fast? Derive the idea of distance per hour.

2. Define speed and write v = d/t; work out one example (120 m in 20 s = 6 m/s).

3. Introduce direction: two cars at 60 km/h towards each other — same speed, different velocities.

4. Define acceleration with a car example: 0 to 20 m/s in 5 s gives a = 4 m/s²; show the formula.

5. Students solve 4 numericals in pairs: two on speed, two on acceleration.

6. Discuss retardation: a bus stopping from 15 m/s in 3 s has a = -5 m/s².

7. Close by summarising the three definitions and their units in a comparison table.

Classroom Activities

• Walk-and-time: students walk 20 m, time it, and compute their own average speed.

• Velocity arrows: pairs draw arrows for cars moving in different directions to show velocity direction.

• Numericals relay: teams take turns solving speed/velocity/acceleration problems on the board.

Assessment Questions

1. Define speed and velocity. How is velocity different from speed?

2. A train covers 240 km in 4 hours. Find its average speed.

3. A car speeds up from 10 m/s to 30 m/s in 8 seconds. Find its acceleration.

4. Why is acceleration called a vector quantity? Give the unit of acceleration.

Homework

• Solve 5 numericals: 2 on average speed, 2 on acceleration, 1 on unit conversion (km/h to m/s).

• Write a short paragraph explaining with an example why a satellite in circular orbit is accelerating.

• Prepare the comparison table of speed, velocity and acceleration with definitions, formulas and units.

Time Allocation

Two periods (80 minutes): 10 min starter question, 15 min speed with example, 15 min velocity and direction, 15 min acceleration with numericals, 15 min activities, 10 min recap and homework.

DM